

The zero-sum sheath stabiliser

ported from SOLPS-ITER `b2stbc_stab_coeff_sheath_te/ti/ni`

The problem. With the floating potential $\Phi = 3T_e$, the wall $E \times B$ drift makes the boundary n_e and T_e overshoot to negative values in 2–3 cells in a *single* time step. Seven different formulations of the drift-corrected parallel-flow condition all failed this way.

The fix. Add to the sheath boundary source of n_e, T_i, T_e :

$$S^{\text{stab}} = \alpha \Pi (X^{\text{old}} - X^{\text{new}}), \quad \Pi = \text{the sheath flux factor } \rho c_s |b_n| \text{ already in the term}$$

In practice: *use* $(\gamma_{\text{sheath}} + \alpha)$ *in the Jacobian*, γ_{sheath} *in the residual*.

Why it is safe:

- **Zero-sum** — vanishes identically when $X^{\text{new}} = X^{\text{old}}$, so the **steady state is unchanged**. It damps only the *approach* to equilibrium, not the answer.
- Pure implicit diagonal: makes the boundary row stiffer against a per-step jump by a factor $1 + \alpha/(\gamma_{\text{sheath}} - 1)$.
- α is **dimensionless** — a multiple of the sheath transmission coefficient, not a physical scale. SOLPS recommends $\alpha = 1-100$; default 0 reproduces previous results bit-for-bit.
- It damps the **transport** rows (n_e, T_i, T_e) , not the momentum condition — SOLPS ships no such term for BCMOM. This matches the observed failure, where n_e and T_e lose positivity together in the same cells.

Result.

- Drift-enabled run: previously died at ≤ 649 steps $\rightarrow$ **now past 1100 and converging**.
- All trends correct: min. wall density flat, $|v_{E,n}|$ falling, and the decrements of min. wall T_e *shrinking* — asymptotic approach, not collapse.

Caveat. SOLPS iterates within a time step, so there the term is purely a convergence aid. JOEREK takes one solve per step, so it also slows genuine transients. The stationary state is still unaffected.